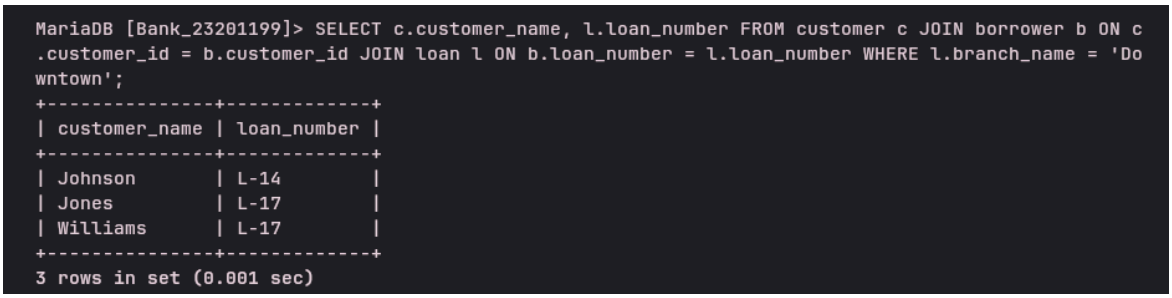
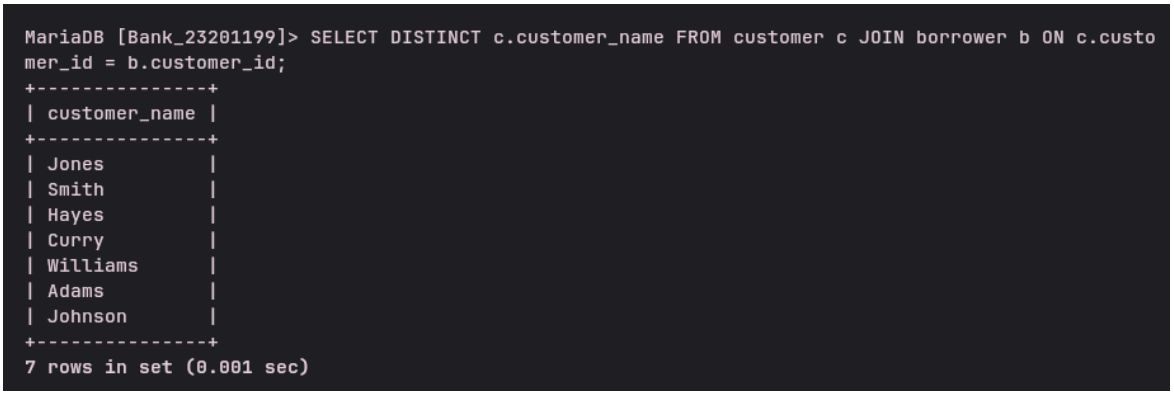
CSE370 : Database Systems

Assignment:03 | Spring 2026

ID : 23201493| Name : Fabiha Tarannum

No 1 Query (as Plain Text)	SELECT c.customer_name, l.loan_number FROM customer c JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number WHERE l.branch_name = 'Downtown';
No 1 SS (of Query & Output in Shell)	
No 2 Query (as Plain Text)	SELECT DISTINCT c.customer_name FROM customer c JOIN borrower b ON c.customer_id = b.customer_id;
No 2 SS (of Query & Output in Shell)	
No 3 Query (as Plain Text)	SELECT customer_name FROM customer WHERE customer_id NOT IN (SELECT customer_id FROM borrower);

No 3 SS (of Query & Output in Shell)

```
MariaDB [Bank_23201199]> SELECT customer_name FROM customer WHERE customer_id NOT IN (SELECT custom
er_id FROM borrower);
+---------------+
| customer_name |
+---------------+
| Lindsay       |
| Turner        |
| Glenn         |
| Brooks        |
| Green         |
+---------------+
5 rows in set (0.001 sec)
```

No 4 Query (as Plain Text)

SELECT c1.customer_name AS Customer1, c2.customer_name AS Customer2, c1.customer_city AS City
FROM customer c1
JOIN customer c2 ON c1.customer_city = c2.customer_city AND c1.customer_id < c2.customer_id;

No 4 SS (of Query & Output in Shell)

```
MariaDB [Bank_23201199]> SELECT c1.customer_name AS Customer1, c2.customer_name AS Customer2, c1.cu
stomer_city AS City FROM customer c1 JOIN customer c2 ON c1.customer_city = c2.customer_city AND c1
.customer_id < c2.customer_id;
+-----------+-----------+------------+
| Customer1 | Customer2 | City       |
+-----------+-----------+------------+
| Jones     | Hayes     | Harrison   |
| Smith     | Curry     | Rye        |
| Lindsay   | Adams     | Pittsfield |
| Turner    | Green     | Stamford   |
+-----------+-----------+------------+
4 rows in set (0.001 sec)
```

No 5 Query (as Plain Text)

SELECT branch_name, ROUND(SUM(balance * 0.04), 2) AS Total_Interest
FROM account
GROUP BY branch_name;

No 5 SS (of Query & Output in Shell)

```
MariaDB [Bank_23201199]> SELECT branch_name, ROUND(SUM(balance * 0.04), 2) AS Total_Interest FROM a
ccount GROUP BY branch_name;
+-------------+----------------+
| branch_name | Total_Interest |
+-------------+----------------+
| Brighton    |          66.00 |
| Downtown    |          20.00 |
| Mianus      |          28.00 |
| Perryridge  |          16.00 |
| Redwood     |          28.00 |
| Round Hill  |          14.00 |
+-------------+----------------+
6 rows in set (0.001 sec)
```

No 6 Query (as Plain Text)	SELECT b.branch_city, a.account_number, a.balance FROM account a JOIN branch b ON a.branch_name = b.branch_name WHERE (b.branch_city, a.balance) IN (SELECT b2.branch_city, MAX(a2.balance) FROM account a2 JOIN branch b2 ON a2.branch_name = b2.branch_name GROUP BY b2.branch_city);
No 6 SS (of Query & Output in Shell)	MariaDB [Bank_23201199]> SELECT b.branch_city, a.account_number, a.balance FROM account a JOIN branch b ON a.branch_name = b.branch_name WHERE (b.branch_city, a.balance) IN (SELECT b2.branch_city, MAX(a2.balance) FROM account a2 JOIN branch b2 ON a2.branch_name = b2.branch_name GROUP BY b2.branch_city); +----------+----------+---------+ | branch_city | account_number | balance | +----------+----------+---------+ | Brooklyn | A-201 | 900 | | Horseneck | A-215 | 700 | | Palo Alto | A-222 | 700 | +----------+----------+---------+ 3 rows in set (0.001 sec)
No 7 Query (as Plain Text)	SELECT * FROM (SELECT l.loan_number, l.amount, c.customer_name FROM loan l JOIN borrower b ON l.loan_number = b.loan_number JOIN customer c ON b.customer_id = c.customer_id ORDER BY l.amount DESC LIMIT 5) AS top_loans ORDER BY amount ASC, loan_number DESC;
No 7 SS (of Query & Output in Shell)	MariaDB [Bank_23201199]> SELECT * FROM (SELECT l.loan_number, l.amount, c.customer_name FROM loan l JOIN borrower b ON l.loan_number = b.loan_number JOIN customer c ON b.customer_id = c.customer_id ORDER BY l.amount DESC LIMIT 5) AS top_loans ORDER BY amount ASC, loan_number DESC; +----------+----------+----------+ | loan_number | amount | customer_name | +----------+----------+----------+ | L-17 | 1000 | Jones | | L-16 | 1300 | Adams | | L-15 | 1500 | Hayes | | L-14 | 1500 | Johnson | | L-23 | 2000 | Smith | +----------+----------+----------+ 5 rows in set (0.001 sec)
No 8 Query (as Plain Text)	SELECT c.customer_name FROM customer c JOIN depositor d ON c.customer_id = d.customer_id JOIN account a ON d.account_number = a.account_number JOIN borrower b ON c.customer_id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number WHERE a.branch_name = 'Perryridge' AND l.branch_name = 'Perryridge';

No 8 SS (of Query & Output in Shell)	

```
MariaDB [Bank_23201199]> SELECT c.customer_name FROM customer c JOIN depositor d ON c.customer_id =
 d.customer_id JOIN account a ON d.account_number = a.account_number JOIN borrower b ON c.customer_
id = b.customer_id JOIN loan l ON b.loan_number = l.loan_number WHERE a.branch_name = 'Perryridge'
AND l.branch_name = 'Perryridge';
+---------------+
| customer_name |
+---------------+
| Hayes         |
+---------------+
1 row in set (0.001 sec)
```

No 9 Query (as Plain Text)	SELECT d.customer_id, SUM(a.balance) AS total_balance FROM depositor d JOIN account a ON d.account_number = a.account_number GROUP BY d.customer_id HAVING COUNT(a.account_number) > 1;
No 9 SS (of Query & Output in Shell)	

```
MariaDB [Bank_23201199]> SELECT d.customer_id, SUM(a.balance) AS total_balance FROM depositor d JOI
N account a ON d.account_number = a.account_number GROUP BY d.customer_id HAVING COUNT(a.account_nu
mber) > 1;
+-------------+---------------+
| customer_id | total_balance |
+-------------+---------------+
| C-226       |          1400 |
+-------------+---------------+
1 row in set (0.001 sec)
```

No 10 Query (as Plain Text)	SELECT d.customer_id, SUM(a.balance) AS total_balance FROM depositor d JOIN account a ON d.account_number = a.account_number GROUP BY d.customer_id HAVING COUNT(a.account_number) > 1;
No 10 SS (of Query & Output in Shell)	

```
MariaDB [Bank_23201199]> SELECT d.customer_id, SUM(a.balance) AS total_balance FROM depositor d JOI
N account a ON d.account_number = a.account_number GROUP BY d.customer_id HAVING COUNT(a.account_nu
mber) > 1;
+-------------+---------------+
| customer_id | total_balance |
+-------------+---------------+
| C-226       |          1400 |
+-------------+---------------+
1 row in set (0.000 sec)
```